

Lab 7

Pedro José

Conexão R -> SQLite

```
library(DBI)
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
con <- dbConnect(SQLite(), "C:/Users/Pedro
José/Downloads/uwmadison.sqlite3")
```

```
dbListTables(con)
```

```
[1] "course_offerings"      "courses"              "grade_distributions"
[4] "instructors"           "rooms"                 "schedules"
[7] "sections"              "subject_memberships"   "subjects"
[10] "teachings"
```

Quem são os professores?

```
SELECT DISTINCT instructors.name
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN sections ON course_offerings.uuid =
sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name
```

Displaying records 1 - 10

name

ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV

name

ALYSSA DIGILIO

Quantos são?

```
SELECT COUNT(DISTINCT instructors.name)
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN sections ON course_offerings.uuid =
sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
```

1 records

COUNT(DISTINCT instructors.name)

377

GPA

```
SELECT course_offerings.name,
       (a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 + c_count*2 +
        d_count*1) * 1.0
       / (a_count + ab_count + b_count + bc_count + c_count + d_count +
        f_count) AS gpa
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN grade_distributions ON course_offerings.uuid =
grade_distributions.course_offering_uuid
WHERE subjects.abbreviation = 'STAT'
```

Displaying records 1 - 10

name	gpa
------	-----

Intro Stats for Engineers	2.959770
Intro Stats for Engineers	3.060976
Intro Stats for Engineers	2.916667
Intro Stats for Engineers	3.000000
Intro Stats for Engineers	2.839286
Intro Stats for Engineers	2.822785

name	gpa
Intro Stats for Engineers	2.919753
Intro Stats for Engineers	3.402299
Intro Stats for Engineers	3.287879
Intro Stats for Engineers	3.591912

Professor Mais Difícil

```
SELECT instructors.name,
       AVG((a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 +
c_count*2 + d_count*1) * 1.0
          / (a_count + ab_count + b_count + bc_count + c_count + d_count
+ f_count)) AS gpa_medio
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN grade_distributions ON course_offerings.uuid =
grade_distributions.course_offering_uuid
INNER JOIN sections ON course_offerings.uuid =
sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
      AND (a_count + ab_count + b_count + bc_count + c_count + d_count +
f_count) > 0
GROUP BY instructors.name
ORDER BY gpa_medio ASC
LIMIT 1
```

1 records

name	gpa_medio
JAMES D KUELBS	2.59875

Professor mais fácil

```
SELECT instructors.name,
       AVG((a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 +
c_count*2 + d_count*1) * 1.0
          / (a_count + ab_count + b_count + bc_count + c_count + d_count
+ f_count)) AS gpa_medio
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN grade_distributions ON course_offerings.uuid =
```

```

grade_distributions.course_offering_uuid
INNER JOIN sections ON course_offerings.uuid =
sections.course_offering_uuid
INNER JOIN teachings ON sections.uuid = teachings.section_uuid
INNER JOIN instructors ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
      AND (a_count + ab_count + b_count + bc_count + c_count + d_count +
f_count) > 0
GROUP BY instructors.name
ORDER BY gpa_medio desc
LIMIT 1;

```

1 records

name	gpa_medio
YAJUAN SI	4

Disciplina mais difícil

```

SELECT course_offerings.name,
      AVG((a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 +
c_count*2 + d_count*1) * 1.0
      / (a_count + ab_count + b_count + bc_count + c_count + d_count
+ f_count)) AS gpa_medio
FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN grade_distributions ON course_offerings.uuid =
grade_distributions.course_offering_uuid
WHERE subjects.abbreviation = 'STAT'
      AND (a_count + ab_count + b_count + bc_count + c_count + d_count +
f_count) > 0
GROUP BY course_offerings.name
ORDER BY gpa_medio ASC
LIMIT 1;

```

1 records

name	gpa_medio
Intro-Theory of Probability	2.891568

Disciplina mais fácil

```

SELECT course_offerings.name,
      AVG((a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 +
c_count*2 + d_count*1) * 1.0
      / (a_count + ab_count + b_count + bc_count + c_count + d_count
+ f_count)) AS gpa_medio

```

```

FROM subjects
INNER JOIN subject_memberships ON subjects.code =
subject_memberships.subject_code
INNER JOIN course_offerings ON subject_memberships.course_offering_uuid =
course_offerings.uuid
INNER JOIN grade_distributions ON course_offerings.uuid =
grade_distributions.course_offering_uuid
WHERE subjects.abbreviation = 'STAT'
AND (a_count + ab_count + b_count + bc_count + c_count + d_count +
f_count) > 0
GROUP BY course_offerings.name
ORDER BY gpa_medio desc
LIMIT 1;

```

1 records

name	gpa_medio
Sample Survey Theory&Method	4

Desconectando

```
dbDisconnect(con)
```

Chunk com Horário

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-22 23:28:07 -03"
```